To address this short coming we instead consider derivations acting on a stalk. Recall the definition of sheaves, stalks, and germs given on *pages 1026-1027*. Then given any C^r manifold M (with $r \in \mathbb{Z}_{\geq 1} \cup \{\infty\}$) and any $0 \leq k \leq r$, we consider the sheaf $\mathcal{F} : \mathbf{Op}(M) \to \mathbf{CAlg}_{\mathbb{R}}$ such that $\mathcal{F}(U) = C^k(U, \mathbb{R})$ and for any open sets $V \subseteq U \subseteq M$ we have that $r_V^U(f) = f|_V$.

Here $C^0(U, \mathbb{R}) \coloneqq C(U, \mathbb{R})$.

For any $p \in M$ we denote the stalk of $\mathcal{F}$ at p to be the set $C_p^k(M)$. Also, given any open set U and C^k map $f : U \to \mathbb{R}$, we typically denote their germ as either $[(U, f)]$ or $[f]$.

Given any germ $[f] \in C_p^k(M)$, it is well-defined to evaluate $f(p)$.

This is because if $[(U, f)] = [(V, g)]$ then there is some open set $W \subseteq U \cap V$ containing p such that $f|_W = g|_W$. But in turn we know that $f(p) = g(p)$.

Now, we say a derivation on $C_p^k(M)$ is an $\mathbb{R}$-linear map $v : C_p^k(M) \to \mathbb{R}$ which also satisfies that if $[f]$ and $[g]$ are germs then $v([f][g]) = f(p)v([g]) + g(p)v([f])$. The set of all derivations on $C_p^k(M)$ is easily checked to be a real vector space and we denote that vector space by $\mathcal{D}_p(M)$.

Let M be an n-dimensional C^r manifold (where $r \in \mathbb{Z}_{\geq 1} \cup \{\infty\}$). Then let $U \subseteq M$ be an open set, let $p \in M$, and consider any smooth chart (V, φ) such that $p \in V \subseteq U$. Note that φ defines local coordinates $(x_1, \ldots, x_n)$ on V. I'll also write $\varphi(p) = (x_1(p), \ldots, x_n(p))$.

Lemma 1: If $f \in C^k(U, \mathbb{R})$ (where $k \geq 1$) and $f(p) = 0$ then there is some open subset $W \subseteq V$ containing p such that:

$$f|_W = \sum_{i=1}^{n} (x_i - x_i(p)) h_i,$$

where $h_i \in C^{k-1}(W, \mathbb{R})$ for all i.

Proof:

To start off, if $\varphi(V)$ is not open in $\mathbb{R}^n$, then by definition $f \circ \varphi^{-1}$ can be extended to being a C^k function $\widetilde{f}$ from an open set $\widetilde{V} \subseteq \mathbb{R}^n$.

Next, let $0 < r$ be a real number such that $\overline{B_r(\varphi(p))} \subseteq \widetilde{V}$. Then for any $x = (x_1, \ldots, x_n) \in \overline{B_r(\varphi(p))}$ we can write:

$$\widetilde{f}(x) = \sum_{i=1}^{n} (x_i - x_i(p)) \int_0^1 \frac{\partial}{\partial x_i} \widetilde{f}(\varphi(p) + s(x - \varphi(p))) \mathrm{d}s$$

By Leibniz's rule, we can conclude that $\widetilde{h}_i : B_r(\varphi(p)) \to \mathbb{R}$ given by:

$$\widetilde{h}_i(x) = \int_0^1 \frac{\partial}{\partial x_i} \widetilde{f}(\varphi(p) + s(x - \varphi(p))) \mathrm{d}s$$

is a C^{k-1} map. And now we let $W = \varphi^{-1}(\varphi(V) \cap B_r(\varphi(p)))$ and $h_i : W \to \mathbb{R}$ be defined by $h_i = \widetilde{h}_i \circ \varphi$. And then each h_i will be a C^{k-1} map and $f|_W = \sum_{i=1}^{n} (x_i - x_i(p)) h_i$. ■

Now let (V, φ) be the smooth chart from before and suppose $f \in C^1(U, \mathbb{R})$ with $f(p) = 0$. If there exists an open set $W \subseteq U \cap V$ such that $f|_W = \sum_{i=1}^n (x_i - x_i(p))h_i$ where each $h_i \in C^0(W)$ and $h_i(\varphi(p)) = 0$ for each i, we say f vanishes to first order at p.

Lemma 2: Let $W, W' \subseteq U \cap V$ both be open sets containing p and then suppose $h_1, \ldots, h_n \in C^0(W, \mathbb{R})$ and $\overline{h_1}, \ldots, \overline{h_n} \in C^0(W', \mathbb{R})$ satisfy that:

$$f|_W = \sum_{i=1}^n (x_i - x_i(p))h_i \text{ and } f|_{W'} = \sum_{i=1}^n (x_i - x_i(p))\overline{h_i}.$$

Then $h_i(p) = \overline{h_i}(p)$ for all i.

Proof:

By considering the set $W \cap W'$ we can assume without loss of generality that $W = W'$. Also, we can assume without loss of generality that W is an open set in $\mathbb{R}^n_{\geq 0}$, H^n, or $\mathbb{R}^n$ since $\overline{h_i}(p) = h_i(p) = 0$ iff $(\overline{h_i} \circ \varphi^{-1})(\varphi(p)) = (h_i \circ \varphi^{-1})(\varphi(p)) = 0$. Finally, note that $f = \sum_{i=1}^n (x_i - x_i(p))h_i = \sum_{i=1}^n (x_i - x_i(p))\overline{h}_i$ means that:

$$0 \equiv \sum_{i=1}^n (x_i - x_i(p)) \cdot (h_i - \overline{h_i}).$$

Thus, we will be done once we show that if $0 \equiv \sum_{i=1}^n (x_i - x_i(p))g_i$ where $g_i \in C^0(W, \mathbb{R})$ then $g_i(x(p)) = 0$ for all i (where $x(p) = (x_1(p), \ldots, x_n(p))$).

Fortunately, consider any $s > 0$ such that $x(p) + se_i \in W$. (Here $e_1, \ldots, e_n$ is the standard basis for $\mathbb{R}^n$). Then $0 = \int_0^s 0 \mathrm{d}t = \int_0^s t g_i(x(p) + te_i)\mathrm{d}t$. But now if $g_i(x(p)) \neq 0$ then by continuity we know there is some $s > 0$ and $\varepsilon > 0$ such that $|g_i(x(0) + te_i)| > \varepsilon$ when $0 \leq t \leq s$. In turn, we have a contradiction as $|\int_0^s t g_i(x(p) + te_i)\mathrm{d}t| \geq s\varepsilon$ means that $0 = \int_0^s t g_i(x(p) + te_i)\mathrm{d}t \neq 0$. ■

More specifically, what the prior lemma shows is that if $f \in C^1(U, \mathbb{R})$ with $f(p) = 0$ and $f|_W = \sum_{i=1}^n (x_i - x_i(p))h_i$ for some open set $W \subseteq U \cap V$ containing p and some C^0 maps $h_1, \ldots, h_n$ then:

$$h_i(p) = \frac{\partial}{\partial x_i}\Big|_p f \text{ for each } i.$$

After all, the above equation holds for the functions $h_1, \ldots, h_n$ constructed in lemma 1, and lemma 2 then says that all other possible $h_1, \ldots, h_n$ must agree with the functions constructed in lemma 1. As a result:

$$f \text{ vanishes to first order iff } \frac{\partial}{\partial x_i}\Big|_p f = 0 \text{ for all } i \text{ iff } v(f) = 0 \text{ for all } v \in T_pU$$

This leads naturally to another result:

Lemma 3: Now suppose (V', ψ) is another smooth chart containing p. We'll let ψ define the local coordinates $(y_1, \ldots, y_n)$ on V'. Also I'll write $\psi(p) = y(p) = (y_1(p), \ldots, y_n(p))$. If $W \subseteq U \cap V$ is an open set containing p and there exists functions $h_i \in C^0(W, \mathbb{R})$ such that $f|_W = \sum_{i=1}^n (x_i - x_i(p))h_i$ and $h_i(p) = 0$ for each i, then for any open set $\Omega \subseteq U \cap V'$ containing p as well as functions $g_i \in C^0(\Omega, \mathbb{R})$ such that $f|_\Omega = \sum_{i=1}^n (y_i - y_i(p))g_i$, we know that $g_i(p) = 0$ for each i.

Proof:
We can conclude via lemmas 1 and 2 that $g_i(p) = \frac{\partial}{\partial y_i}\Big|_p f$ for each i. But now note from *page 1052* that each linear operator $\frac{\partial}{\partial y_i}\Big|_p$ is a linear combination of the operators $\frac{\partial}{\partial x_j}\Big|_p$. And since $\frac{\partial}{\partial x_j}\Big|_p f = h_j(p) = 0$ for each j by assumption, we can conclude $g_i(p) = 0$ for each i. ■

As a result of the prior lemmas, we can say that f vanishing to first order at p is independent of our chosen local coordinates or the particular functions in our expression:

$$f|_W = \sum_{i=1}^{n}(x_i - x_i(p))h_i.$$

Also as is hopefully obvious, if $[(U, f)] = [(\widetilde{U}, \widetilde{f})]$ in $C^1_p(M)$ and f vanishes to first order at p then so does $\widetilde{f}$. So, we can talk about a germ in $C^1_p(M)$ vanishing to first order.

Now here's the new and improved definition of the tangent plane of p. For now I'll denote my old construction as T_pM. Meanwhile, I'll define a new construction by:

$$\mathcal{T}_pM \coloneqq \{v \in \mathcal{D}_p(M) : v([f]) = 0 \text{ if } [f] \text{ vanishes to first order}\}.$$

Clearly $\mathcal{T}_pM$ is a vector subspace of $\mathcal{D}_p(M)$.

Theorem: There is a bijective linear map $\Phi : T_pM \to \mathcal{T}_pM$ such that $(\Phi(v))([f]) \coloneqq v(f)$ for all $f \in C^k(M, \mathbb{R})$.

Proof:
Suppose $[(U, f)]$ is any germ and note that by multiplying on a C^r bump function, we can find a function $g \in C^1(M, \mathbb{R})$ such that $[(M, g)] = [(U, f)]$. So, for any $v \in T_pM$ we let $(\Phi(v))([f]) \coloneqq v(g)$.

To see that Φ is well-defined, note that if $[(M, g)] = [(U, f)] = [(M, \widetilde{g})]$ then $g = f = \widetilde{g}$ on a neighborhood of p. So by proposition 3.8 on *page 1033* we know that $v(g) = v(\widetilde{g})$.

Showing that $\Phi(v) \in \mathcal{D}_p(M)$ is then easy. Furthermore, if $[(U, f)]$ vanishes to first order then we know for any $g \in C^k(M, \mathbb{R})$ with $[(M, g)] = [(U, f)]$ that $v(g) = 0$. So, $\Phi(v) \in \mathcal{T}_pM$. Finally, it is easy to verify that $\Phi(c_1v_1 + c_2v_2) = c_1\Phi(v_1) + c_2\Phi(v_2)$ for all $c_1, c_2 \in \mathbb{R}$ and $v_1, v_2 \in T_pM$. So, Φ is a linear map from T_pM to $\mathcal{T}_pM$.

To show that Φ is an isomorphism, let (U, φ) be a smooth chart containing p and defining local coordiantes $\varphi(m) = x(m) = (x_1(m), \ldots, x_n(m))$. Then recall $\frac{\partial}{\partial x_1}\Big|_p, \ldots, \frac{\partial}{\partial x_n}\Big|_p$ is a basis for T_pM. Meanwhile, we can define a basis for $\mathcal{T}_pM$ as follows:

Let $[(V, f)] \in C^1_p(M)$. By lemma 1 from a few pages ago can find an open subset $W \subseteq V \cap U$ as well as functions $h_i \in C^0(W, \mathbb{R})$ such that:

$$f|_W = f(p) + (x_i - x_i(p))h_i.$$

And now we define $\omega_i([f]) \coloneqq h_i(p)$. This is well-defined because if $[(V,f)] = [(\widetilde{V},\widetilde{f})]$ then we know there is some open set $\Omega \subseteq V \cap \widetilde{V}$ such that $f|_\Omega = \widetilde{f}|_\Omega$. And now upon writing that $f|_W = f(p) + \sum_{i=1}^n (x_i - x_i(p))h_i$ and $\widetilde{f}|_{\widetilde{W}} = f(p) + \sum_{i=1}^n (x_i - x_i(p))\widetilde{h}_i$, we know that:
$(f - f(p))|_{\Omega\cap\widetilde{W}} = \sum_{i=1}^n (x_i - x_i(p))\widetilde{h}_i$ and $(f - f(p))|_W = \sum_{i=1}^n (x_i - x_i(p))h_i$.

So by lemma 2 from before, we know that $h_i(p) = \widetilde{h}_i(p)$ for each i. And this proves that $\omega_i([f]) = \omega_i([\widetilde{f}])$.

Showing that $\omega_i \in \mathcal{T}_pM$ is now easy.

To show one necessary algebraic property of ω_i suppose $[f],[g] \in C_p^k(M)$ and write:
$f|_W = f(p) + \sum_{i=1}^n (x_i - x_i(p))\widetilde{h}_i$ and $g|_{\widetilde{W}} = g(p) + \sum_{i=1}^n (x_i - x_i(p))\widetilde{h}_i$.

Then after regrouping terms we can see that:
$$(fg)|_{W\cap\widetilde{W}} = (fg)(p) + \sum_{i=1}^n (x_i - x_i(p)) \cdot \left(h_i g + \widetilde{h}_i f - (x_i - x_i(p))h_i\widetilde{h}_i\right)$$

So, $\omega_i([f][g]) = \omega_i([f])g(p) + \omega_i([f])f(p) - 0$.

I also claim $\omega_1,\ldots,\omega_n$ span $\mathcal{T}_pM$. After all, let $\nu \in \mathcal{T}_pM$ be an arbitrary derivation.
We can consider the germ $[(U,f_i)]$ such that $f_i(m) = x_i(m) - x_i(p)$.
And then we let $c_i = \nu([f_i])$. Now for any germ $[f] \in C_p^k$, upon writing $f = f(p) + \sum_{i=1}^n (x_i - x_i(p))h_i$ we know that:
$f - f(p) - \sum_{i=1}^n (x_i - x_i(p))h_i(p)$ vanishes to first order at p.

So:
$$\begin{aligned} 0 &= \nu\left([f - f(p) - \textstyle\sum_{i=1}^n (x_i - x_i(p))h_i(p)]\right) \\ &= \nu([f]) - f(p)\nu([0]) - \textstyle\sum_{i=1}^n \nu([x_i - x_i(p)])h_i(p) \\ &= \nu([f]) - 0 - \textstyle\sum_{i=1}^n c_i\omega_i([f]). \end{aligned}$$

Or in other words, $\nu = \sum_{i=1}^n c_i\omega_i$.

And finally, I claim that $\omega_1,\ldots,\omega_n$ are all linearly independent. After all, suppose $c_1,\ldots,c_n \in \mathbb{R}$ satisfy that $\sum_{i=1}^n c_i\omega_i([f]) = 0$ for every germ $[f]$. Then in particular we know that $c_j = \sum_{i=1}^n c_i\omega_i([x_j - x_j(p)]) = 0$ for every j.

And now to finish the proof, just note for any germ $[(U,f)]$ that:
$$\Phi\left(\left.\frac{\partial}{\partial x_i}\right|_p\right)([f]) = \left.\frac{\partial}{\partial x_i}\right|_p f = \omega_i([f]) \text{ for each } i. \blacksquare$$

For the sake of convenience, I'll identify ω_i from the prior theorem with $\left.\frac{\partial}{\partial x_i}\right|_p$.

One last thing to sort out is how to define the differential of a C^1 map $F : M \to N$ with respect to the vector space $\mathcal{T}_pM$. To do this, we merely say:
$(\mathrm{d}F_p(\nu))([(U,f)]) \coloneqq \nu([(F^{-1}(U), f\circ F)])$ for every $\nu \in \mathcal{T}_pM$ and $[(U,f)] \in C^1_{F(p)}(N)$.

Checking that $\mathrm{d}F_p(\nu)$ is a well-defined operator in $\mathcal{D}_{F(p)}(N)$ for every $\nu \in \mathcal{T}_pM$ and that $\mathrm{d}F_p : \mathcal{T}_pM \to \mathcal{D}_{F(p)}(N)$ is a well-defined linear map is easy.

As for proving that $\mathrm{d}F_p(\nu)$ annihilates all germs vanishing to first order, it suffices to prove that if $[f] \in C^1_{F(p)}(N)$ vanishes to first order then so does $[f \circ F] \in C^1_p(M)$.

To do this, let $[(V, f)]$ be a germ and let $\widetilde{\mathrm{d}}F_p : T_p(F^{-1}(V)) \to T_{F(p)}V$ be the old definition of the differential of F. Then note that $f \circ F$ vanishes to first order iff $v(f \circ F) = 0$ for all $v \in T_p(F^{-1}(V))$. Yet simultaneously $v(f \circ F) = (\mathrm{d}F_p(v))(f)$. And since f vanishes to first order and $\mathrm{d}F_p(v) \in T_{F(p)}V$, we know that:

$$(\mathrm{d}F_p(v))(f) = 0.$$

Also if $\Phi_M : T_pM \to \mathcal{T}_pM$ and $\Phi_N : T_{F(p)}N \to \mathcal{T}_{F(p)}N$ are the maps constructed in the prior theorem, then we can see by looking at a representive for each germ whose domain is all of M or N that:

$$\begin{array}{ccc} & f \mapsto v(f) \qquad\qquad g \mapsto v(g \circ F) & \\ T_pM & \xrightarrow{\widetilde{\mathrm{d}}F_p} & T_{F(p)}N \\ \wr \downarrow \Phi_M & & \wr \downarrow \Phi_N \\ \mathcal{T}_pM & \xrightarrow{\mathrm{d}F_p} & \mathcal{T}_{F(p)}N \\ & [f] \mapsto v(f) \qquad\qquad [g] \mapsto v(g \circ F) & \end{array}$$

Anyways this new definition of the tangent plane is the most general way the tangent plane of a manifold is defined. It is also nice because as mentioned before, it generalizes to types of manifolds (i.e. real-analytic and complex) where germs don't have representatives in any arbitrary neighborhood of a point.

8/17/2026

I'm going to keep being distracted due to the fact I've been in and out of being homeless for two weeks. But right now I kinda want to study again. I'll be studying Lee's book again where I left off.

Local Immersion Theorem For Manifolds with Boundaries: Let M be an m-dimensional C^r manifold with a boundary (but no corners), and let N be an n-dimensional C^r manifold without a boundary. Also suppose $F : M \to N$ is a C^r immersion. For any $p \in \partial M$ there exists a smooth boundary chart (U, φ) on M containing p as well as a smooth chart (V, ψ) on N such that $\varphi(p) = 0$, $F(U) \subseteq V$, and $\widehat{F} := \psi \circ F \circ \varphi^{-1}$ is given by:

$$\widehat{F}(x_1, \ldots, x_m) = (x_1, \ldots, x_m, \underbrace{0, \ldots, 0}_{n-m \text{ zeros}})$$

Proof:
By choosing local coordinates around p it suffices to assume M is an open subset of H^m and N is an open set $\mathbb{R}^n$. This follows by identical reasoning as in the rank theorem. Also by identical reasoning as in the rank theorem, we can assume $p = 0$ and $F(p) = 0$.

Next we know that F extends to a C^r function $\widetilde{F} : W \to \mathbb{R}^n$ such that $M \subseteq W$ and $\widetilde{F}|_M = F$. Because $\mathrm{d}\widetilde{F}_0 = \mathrm{d}F_0$ is an injection (here we recall that we can identify T_pM with T_pW), we can shrink W and M such that $\widetilde{F}$ is a C^r immersion. Also, we can shrink W (without shrinking M) such that $\widetilde{F}(W) \subseteq N$. And now let us write the coordinates on W as $(x_1, \ldots, x_m)$ and the coordinates on N as $((v_1, \ldots, v_m), (w_1, \ldots, w_{n-m}))$.

By the rank theorem (see *page 1062*), there exists smooth charts (U_0, φ_0) and (V_0, ψ_0) such that $0 \in U_0 \subseteq W$, $F(U_0) \subseteq V_0 \subseteq N$, $\varphi_0(0) = 0$, and $\widehat{F} \coloneqq \psi_0 \circ \widetilde{F} \circ \varphi_0^{-1}$ is given by:

$$\widehat{F}(x_1, \ldots, x_m) = (x_1, \ldots, x_m, \underbrace{0, \ldots, 0}_{n-m \text{ zeros}})$$

The one issue that needs to be worked out is that $\varphi_0(U_0 \cap M)$ may not be contained in H^m. To address this, we make the following modification:

φ_0 is a C^r diffeomorphism from $U_0 \subseteq \mathbb{R}^m$ to an open set $\widehat{U}_0 \subseteq \mathbb{R}^m$ such that $\varphi_0(0) = 0$. In turn, $\varphi_0^{-1} \times \mathrm{Id}_{\mathbb{R}^{n-m}} : \widehat{U}_0 \times \mathbb{R}^{n-m} \to U_0 \times \mathbb{R}^{n-m}$ is a C^r diffeomorphism defined on an open neighborhood of 0.

Next let $\psi \coloneqq (\varphi_0^{-1} \times \mathrm{Id}_{\mathbb{R}^{n-m}}) \circ \psi_0$. This is well-defined C^r diffeomorphism from an open set $V \coloneqq V_0 \cap \psi_0^{-1}(\widehat{U}_0 \times \mathbb{R}_{n-m})$ of N containing $F(0) = 0$.

Finally $U \coloneqq U_0 \cap F^{-1}(V) \cap M$ is an open subset of M containing 0 such that for all $x \in U$:

$$\begin{aligned}\psi \circ F(x_1, \ldots, x_m) &= (\varphi_0^{-1} \times \mathrm{Id}_{\mathbb{R}^{n-m}}) \circ \psi_0 \circ F \circ \varphi_0^{-1} \circ \varphi_0(x_1, \ldots, x_m) \\ &= (\varphi_0^{-1} \times \mathrm{Id}_{\mathbb{R}^{n-m}})(\varphi(x_1, \ldots, x_m), 0) = (x_1, \ldots, x_m, 0, \ldots, 0)\end{aligned}$$

Therefore the original coordinates for M restricted to U as well as the smooth chart (V, ψ) works. ■

Given C^r manifolds M and N, we say a map $F : M \to N$ is a C^k embedding (where $k \leq r$) if F is a C^k immersion and F is a homeomorphism from M to $F(M)$.

Exercise 4.16: If M, N, P are C^r manifolds, $k \leq r$, and $F : M \to N$ and $G : N \to P$ are C^k embeddings then $(G \circ F) : M \to P$ is a C^k embedding as well.

Proof:
$G \circ F$ is obviously C^k. Also a composition of immersions will be an immersion. After all, for all $p \in M$ we have that $\mathrm{d}(G \circ F)_p = \mathrm{d}G_{F(p)} \circ \mathrm{d}F_p$ is injective since $\mathrm{d}F_p$ and $\mathrm{d}G_{F(p)}$ are both injective. Finally, let $h_F : F(M) \to M$ and $h_G : G(N) \to N$ be the continuous inverse functions to F and G respectively. Then $(h_F \circ h_G)|_{G(F(M))}$ is a continuous inverse function to $G \circ F$. ■

Some examples of C^r embeddings are:

1. If M is a C^r manifold and $U \subseteq M$ is an open set, then the inclusion map $\iota : U \hookrightarrow M$ is a C^r embedding.

   It's C^r, at every point it's differential is just the identity linear map, and also it's injective with an inverse map from $\iota(U) \to U$ just being the identity function (which is obviously continuous).

2. Suppose $M_1, \ldots, M_k$ are C^r manifolds and $p_i \in M_i$ is an arbitrarily chosen point for each i. Then for every $1 \leq j \leq k$ we have that the map $\iota_j : M_j \to M_1 \times \cdots \times M_k$ given by $\iota_j(m_j) = (p_1, \ldots, p_{j-1}, m_j, p_{j+1}, \ldots, p_k)$ is a C^r embedding.

   That ι_j is C^r is immediate. Also we know that it has a continuous C^r inverse $\pi_j : M_1 \times \cdots \times M_k \to M_j$ given by $(m_1, \ldots, m_{j-1}, m_j, m_{j+1}, \ldots, m_k) \mapsto m_j$. Finally, for any $q \in M_j$ we have that$\mathrm{d}(\mathrm{Id}_{M_j})_q = \mathrm{d}\pi_{\iota_j(q)} \circ \mathrm{d}(\iota_j)_q$ implies that $\mathrm{d}\pi_{\iota_j(q)}$ is surjective and $\mathrm{d}(\iota_j)_q$ is injective. From this we can conclude that ι_j is an immersion and π_j is a submersion.

Next, here are two nonexamples of smooth embeddings:

1. Consider the map $\gamma : \mathbb{R} \to \mathbb{R}^2$ given by $\gamma(t) = (t^3, 0)$. This is a C^∞ map with a continuous inverse function given by $(t, 0) \mapsto (\sqrt[3]{t}, 0)$. That said, $\mathrm{d}\gamma_0$ is a trivial linear map since $\gamma'(0) = 0$. So, γ fails to be an immersion.

2. Consider the lemniscate $\beta : (-\pi, \pi) \to \mathbb{R}^2$ defined by $\beta(t) = (\sin(2t), \sin(t))$.

   Proving that β is injective just involves considering a bunch of subcases. So suppose $s, t \in (-\pi, \pi)$ satisfy that $\sin(2t) = \sin(2s)$ and $\sin(t) = \sin(s)$.

   - If $t \in (0, \pi/2)$ and $s \neq t$ then $\sin(t) = \sin(s) \implies s = \pi - t$. Yet then $\sin(2s) < 0$ while $\sin(2t) > 0$. So if $t \in (0, \pi/2)$ then $s = t$.
   - If $t \in (\pi/2, \pi)$ and $s \neq t$ then $\sin(t) = \sin(s) \implies s = \pi - t$ again. Yet now we have that $\sin(2t) < 0$ while $\sin(2s) > 0$. So if $t \in (\pi/2, \pi)$ then $s = t$.
   - Given that $t \in (-\pi, \pi)$, we have that $\sin(t) = 0$ iff $t = 0$ and $\sin(t) = 1$ iff $t = \pi/2$. So, β is injective on $[0, \pi)$.
   - By the oddness of $\sin(2t)$ and $\sin(t)$ we can also conclude that β is injective on $(-\pi, 0]$. And since $\sin(t) < 0$ for all $t < 0$ and $\sin(t) > 0$ for all $t > 0$, we can conclude that β is injective on $(-\pi, \pi)$.

   We can also see that β is a C^∞ map. In particular, $\beta'(t) = (2\cos(2t), \cos(t))$. But now $\cos(t) = 0$ iff $t = \pm\pi/2$. Yet $\cos(2(\pi/2)) = -1 = \cos(2(-\pi/2))$. So $\beta'(t) \neq 0$ for any $t \in (-\pi, \pi)$ and we can conclude that β is an immersion.

   That all said, β does not have a continuous inverse function. After all, $\beta(0) = 0$ but also $\beta(t) \to 0$ as $t \to \pi$ or as $t \to -\pi$. Therefore, if $f : \beta((-\pi, \pi)) \to (-\pi, \pi)$ was the inverse function to β then we'd have to have that f is not continuous at 0.

Proposition 4.22: Suppose M and N are C^r manifolds with or without boundary, and $F : M \to N$ is an injective C^k immersion (where $k \leq r$). Then F is a C^k embedding if any of the following conditions hold:

(a) F is an open or closed map into N.

Why?
F being an open map into N trivially means F is an open map into the subspace topology of $F(M) \subseteq N$.

Meanwhile, if F is a closed map then let $U \subseteq M$ be an open set. Then $M - U$ is a closed set so $F(M - U)$ is a closed subset of N. It follows that $N - F(M - U)$ is an open subset of N. And so $F(M) \cap (N - F(M - U)) = F(U)$ (which holds because F is injective) is an open set in the subspace topology of $F(M) \subseteq N$.

More generally, this reasoning shows that if X and Y are topological spaces then any injective continuous map $f : X \to Y$ that is also open or closed is a topological embedding (i.e. a homeomorphism from X to $f(X) \subseteq Y$).

(b) F is a proper map.

Why?
Let E be any closed subset of M and then suppose q is any accumulation point of $F(E)$ in N. Since N is an LCH space, we can pick a precompact open set $V \subseteq N$ containing q. And now $\overline{V} \cap F(E)$ also has q as an accumulation point. Yet because F is proper, we know that $F^{-1}(\overline{V})$ is compact. Then since closed subsets of compact sets are also compact, we know that $E \cap F^{-1}(\overline{V})$ is compact. In turn, as F is continuous we have that $F(E \cap F^{-1}(\overline{V})) = F(E) \cap \overline{V}$ is compact in N (and thus closed since N is Hausdorff). So $F(E) \cap \overline{V}$ must contain all its accumulation points and so $q \in F(E)$.

Since q was arbitrary, we've proven that $F(E)$ is a closed subset of N. So, F is a closed map into N and we can finish by citing part (a).

More generally, this reasoning shows that if X is a topological space and Y is an LCH space then any continuous proper map $f : X \to Y$ is also a closed map.

(c) M is a compact manifold.

Why?
$F : M \to F(N)$ is a continuous bijection, M is compact, and $F(N)$ is Hausdorff. Thus F is a homeomorphism.

(d) M has no boundary and $\dim(M) = \dim(N)$.

Why?
By proposition 4.8(b) on *page 1061*, we know F is a local C^k-diffeomorphism. It then follows by proposition 4.6(c) on the page before proposition 4.8(b) that F is an open map. So, we finish by invoking part (a) of this theorem. ■

8/20/2026

I was realizing yesterday that I've conflated open maps with closed maps an unhealthy amount of times in my brain and am not sure if that's made some of my prior notes incorrect. I've double checked a bunch of pages and not found any errors but still.

Anyways a thing I want to get across first and foremost today is that a closed surjective continuous map is a quotient map for different reasons then an open surjective continuous map is.

Proposition: Suppose X and Y are topological spaces and $f : X \to Y$ is a continuous surjective map. Then f is a quotient map if either of the following holds:

(a) f is an open map.

Why?
Suppose $U \subseteq Y$ satisfies that $f^{-1}(U)$ is an open subset of X. Then because f is surjective, we know that $f(f^{-1}(U)) = U$. And because f is open, we know that U is open.

(b) f is a closed map.

Why?
Suppose $U \subseteq Y$ again satisfies that $f^{-1}(U)$ is an open subset of X. Then we know that $X - f^{-1}(U)$ is a closed subset of X. So because f is a closed map, we know that $f(X - f^{-1}(U)) = f(X) - U$ is a closed subset of Y. Since f is surjective, we in other words know that $Y - U$ is a closed set. Equivalently, U is an open set. ■

Another thing I want to mention is that I should think of a C^k embedding as the differential-topology equivalent of a topological embedding. Similarly, a (surjective) C^k submersion is the differential-topology equivalent of a quotient map as I will soon show.

One final observation I want to make before getting back into taking notes on Lee's book is that we can precisely restate the definition of a coordinate patch α for a k-dimensional C^r manifold in $\mathbb{R}^n$ given on *page 84* as being that α is a C^r-embedding. I will come back to this soon when talking about embedded submanifolds.

Theorem 4.25 (Local Embedding Theorem): Suppose M and N are C^r manifolds where M has no corners and $F : M \to N$ is a C^k map. Then F is a C^k immersion if and only if for all $p \in M$ there exists an open set $U \subseteq M$ such that $F|_U : U \to N$ is a C^k embedding.

$(\Longleftarrow)$
This direction is immediate.

$(\Longrightarrow)$
Suppose F is a C^k immersion. Without loss of generality we can assume M and N are equipped with the unique C^k structures containing their original C^r structures since that would have no effect on whether F is a C^k immersion or $F|_U$ is a C^k embedding. And this assumption is useful cause it let's us apply the rank theorem.

Firstly we show that any $p \in M$ has a neighborhood U on which F is injective.

First suppose $F(p) \notin \partial N$. Then as the set of interior points $\text{Int}(N)$ is an open set and F is continuous, we know that there is an open set $\Omega \subseteq M$ containing p with $F(\Omega) \subseteq \text{Int}(N)$. Then by either the rank theorem on *page 1062* or the local immersion theorem on *page 1101*, we know there is an open set $U_1 \subseteq M$ of p where F has the coordinate representation:

$$(x_1, \ldots, x_m) \mapsto (x_1, \ldots, x_m, 0, \ldots, 0).$$

So F is injective on U_1.

Meanwhile, suppose $F(p) \in \partial N$ and let (W, ψ) be a smooth chart on N containing $F(p)$. Then $U_0 := F^{-1}(W)$ is an open subset of M containing p. Furthermore, let $\iota : H^n (\text{ or } \mathbb{R}^n_{\geq 0}) \hookrightarrow \mathbb{R}^n$ be the inclusion map (here $n = \dim(N)$). Then $\iota \circ \psi \circ F|_{U_0}$ is a C^k map from M to a C^k manifold without boundary. Also, as $\mathrm{d}\iota_a$ and $\mathrm{d}\psi_b$ are bijections for all $a \in H^n$ and $b \in W$, we can conclude that $\iota \circ \psi \circ F|_{U_0}$ is a C^k-immersion still. Thus, we can apply the earlier reasoning to this map to find an open set $U_1 \subseteq U_0$ containing p such that $(\iota \circ \psi \circ F|_{U_0})|_{U_1}$ is an injection. Equivalently, this implies $F|_{U_1}$ is an injection.

Now consider any $p \in M$ and let $U_1 \subseteq M$ be an open set containing p on which F is injective. Since M is an LCH space, we can find a precompact open set U such that $p \in U \subseteq \overline{U} \subseteq U_1$. But now $F|_{\overline{U}}$ is an injective continuous function from a compact set. Therefore, $F|_{\overline{U}}$ is a topological embedding. Next since restrictions of topological embeddings are still topological embeddings, we know $F|_U$ is a topological embedding. And since $F|_U$ is a C^k immersion, we can conclude that $F|_U : U \to N$ is a C^k embedding. ■

Suppose M and N are C^r manifolds. If $\pi : M \to N$ is a continuous map then a section for π is a continuous right inverse function $\sigma : N \to M$ (i.e. $\pi \circ \sigma = \text{Id}_N$). A local section for π is a continuous map $\sigma : U \to M$ where $U \subseteq N$ is an open set and $\pi \circ \sigma = \text{Id}_U$.

Theorem 4.26 (Local Section Theorem): Suppose M and N are C^r manifolds without boundaries, and $\pi : M \to N$ is a C^k map (where $k \leq r$). Then π is a C^k submersion if and only if every $p \in M$ is in the image of a C^k local section of π.

$(\Longleftarrow)$

Given any $p \in M$, let $\sigma : U \to M$ be a C^k local section such that $q \in U$ and $\sigma(q) = p$. In other words, $q = \pi(\sigma(q)) = \pi(p)$. But now since $\pi \circ \sigma = \text{Id}_U$, we can conclude that $\mathrm{d}\pi_p \circ \mathrm{d}\sigma_q = \text{Id}_{T_q N}$. In turn, we must have that $\mathrm{d}\pi_p$ is surjective. And since p was arbitrary, we can conclude π is a submersion.

$(\Longrightarrow)$

Suppose π is a C^k submersion. Then without loss of generality we can assume M and N are equipped with the unique C^k structures containing their original C^r structures. After all, this change won't effect whether π is a C^k submersion, or whether some local section of π is C^k or not. Also, this assumption lets us apply the rank theorem.

Pick any $p \in M$ and let $q = \pi(p)$. By the rank theorem, we can choose smooth charts (V_0, φ) on M containing p and (U_0, ψ) containing q such that $\varphi(p) = 0$, $\pi(V_0) \subseteq U_0$, and $\widehat{\pi} := \psi \circ \pi \circ \varphi^{-1}$ has the form $\widehat{\pi}(x_1, \ldots, x_n, x_{n+1}, \ldots, x_m) = (x_1, \ldots, x_n)$.

For a small enough $\varepsilon > 0$ we have that the coordinate cube
$$C_\varepsilon := \{(x_1, \ldots, x_m) : |x_i| < \varepsilon \text{ for all } i\}$$
is an open subset of $\varphi(V_0)$ containing $0 = \varphi(p)$. In turn we have that:
$$C'_\varepsilon := \{(y_1, \ldots, y_n) : |y_i| < \varepsilon \text{ for all } i\}$$
is an open subset of $\psi(U_0)$ containing $0 = \widehat{\pi}(0) = \widehat{\pi} \circ \varphi(p) = \psi(q)$ and satisfying that $\widehat{\pi}(C_\varepsilon) = C'_\varepsilon$. Let $U = \psi^{-1}(C'_\varepsilon)$. Then define $\iota : C'_\varepsilon \to C_\varepsilon$ by:
$$\iota(y_1, \ldots, y_n) = (y_1, \ldots, y_n, \underbrace{0, \ldots, 0}_{m-n \text{ zeros}}).$$

Finally, let $\sigma = \varphi^{-1} \circ \iota \circ \psi|_U$. Then if $q' \in U$ and $\varphi(q') = (y_1, \ldots, y_n)$ we have that:
$$\begin{aligned} \pi \circ \sigma(q') &= \psi^{-1} \circ \widehat{\pi} \circ \varphi \circ (\varphi^{-1} \circ \iota \circ \psi)(q') \\ &= \psi^{-1} \circ \widehat{\pi} \circ \iota(y_1, \ldots, y_n) = \psi^{-1} \circ \widehat{\pi}(y_1, \ldots, y_n, 0, \ldots, 0) \\ &= \psi^{-1}(y_1, \ldots, y_n) = q' \end{aligned}$$

So $\pi \circ \sigma = \mathrm{Id}_U$. Also σ is a C^k map and
$$\sigma(q) = \varphi^{-1} \circ \iota(0, \ldots, 0) = \varphi^{-1}(0, \ldots, 0) = p. \blacksquare$$

Based on theorems 4.25 and 4.26, we also get the following purely topological definitions. Suppose X and Y are topological spaces and $f : X \to Y$ is a continuous function.

- f is a topological immersion if every point $x \in X$ has a neighborhood $U \subseteq X$ such that $f|_U : U \to Y$ is a topological embedding.
- f is a topological submersion if every point $x \in X$ is in the image of a (continuous) local section for f.

The prior two theorems say that any C^k immersion / submersion is a topological immersion / submersion. That said, be aware that a topological immersion or submersion that happens to also be a C^k map is not necessarily a "C^k immersion" or "C^k submersion".

For example, let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(t) = t^3$. Then f is a C^∞ map which is also a homeomorphism. And clearly homeomorphisms are trivially topological immersions and submersions. That said, as $\mathrm{d}f_0$ is the trivial linear map as $f'(0) = 0$, we know f is not a C^∞ immersion or submersion.

Theorem 4.27: Let M and N be C^r manifolds without boundaries, and suppose $\pi : M \to N$ is a C^k submersion. Then π is an open map. Also consequently, if π is surjective then π is a quotient map.

Proof:
Suppose $W \subseteq M$ is an open set and $q \in \pi(W) \subseteq N$. We can pick $p \in W$ with $\pi(p) = q$. And then by the local section theorem, we know there is an open subset $U \subseteq N$ containing q as well as a C^k map $\sigma : U \to M$ such that $\sigma(q) = p$ and $\pi \circ \sigma = \mathrm{Id}_U$. Yet now note

that $\sigma^{-1}(W)$ is an open subset of U (since σ is continuous) and it contains q. Also note that if $y \in \sigma^{-1}(W)$ then $y = \pi(\sigma(y)) \in \pi(W)$ Therefore, we've shown that q is in the topological interior of $\pi(W)$. And since q was arbitrary, this proves that $\pi(W)$ is an open set. ■

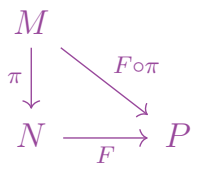
Exercise 4-6: Let M be a nonempty compact C^r manifold without a boundary, and let N be a connected non-compact C^r without a boundary. Then there is no C^k submersion $\pi : M \to N$ (where $k \leq r$).

Proof:
By the prior theorem we must have that such a function π is an open map. Thus in particular we can conclude that $\pi(M)$ is an open subset of N. That said, as M is compact and π is continuous, we must also have that $\pi(M)$ is a compact subset of N. Since N is Hausdorff, we can conclude $\pi(M)$ is a clopen set. And since N is connected, we thus must have that $\pi(M)$ is either the empty set or all of N.

$\pi(M)$ can't be the empty set however because $M \neq \emptyset$. Simultaneously, since N is not compact we can't have that $\pi(M) = N$. Thus we have a contradiction. ■

The next theorem is a generalization of exercise 4.10(b) on *pages 1061-1062*.

Theorem 4.29: Suppose M and N are C^r manifolds without boundaries and $\pi : M \to N$ is a surjective C^k submersion. For any C^r manifold P we have that any map $F : N \to P$ is C^k iff $F \circ \pi : M \to P$ is a C^k map.



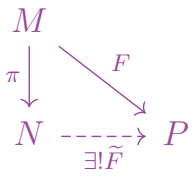
Proof:
If F is C^k then the composition $F \circ \pi$ will also be C^k. So, one direction of this theorem is trivial.

Conversely, suppose $F \circ \pi$ is C^k and let $q \in N$ be any point. Then as π is surjective we know that there exists $p \in M$ with $\pi(p) = q$. By the local section theorem, we know there exists an open set $U \subseteq N$ containing q as well as a C^k map $\sigma : U \to M$ such that $\sigma(q) = p$ and $\pi \circ \sigma = \mathrm{Id}_U$. And now $F|_U = F \circ \mathrm{Id}_U = F \circ (\pi \circ \sigma) = (F \circ \pi) \circ \sigma$ is a C^k map.

Since every point in N has a neighborhood on which F is a C^k map, we can conclude that F is globally a C^k map. ■

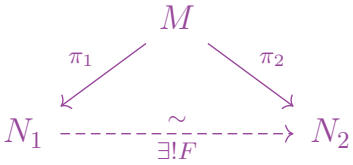
Corollary 4.30: Suppose M and N are C^r manifolds without boundaries and $\pi : M \to N$ is a surjective C^k submersion. If P is any C^r manifold and $F : M \to P$ is any C^k map that is constant on the fibers of π (meaning F is constant on each set $\pi^{-1}(q)$ for every $q \in N$),

then there exists a unique C^k map $\widetilde{F} : N \to P$ such that $F = \widetilde{F} \circ \pi$.

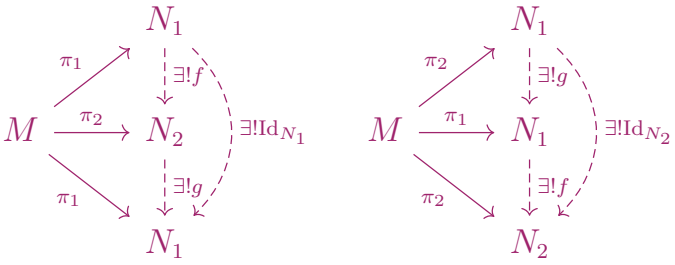


Proof:
Since π is a quotient map, we know from the universal property of quotient maps that there exists a unique continuous function $\widetilde{F} : N \to P$ such that $F = \widetilde{F} \circ \pi$. Then as F is C^k, the prior theorem tells us that $\widetilde{F}$ is also C^k. ■

Corollary 4.31: Suppose M, N_1, N_2 are C^r manifolds without boundaries and $\pi_1 : M \to N_1$ and $\pi_2 : M \to N_2$ are surjective C^k submersions that are constant on each other's fibers (i.e. $\pi_1(x) = \pi_1(y) \iff \pi_2(x) = \pi_2(y)$). Then there exists a unique C^k diffeomorphism $F : N_1 \to N_2$ such that $\pi_2 = F \circ \pi_1$.



Why?
Just consider the following two diagrams:



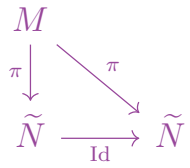
Something I want to quickly mention is that an anlogous corollary can be proven for quotient maps $p_1 : X \to Y_1$ and $p_2 : X \to Y_2$ (where X, Y_1, Y_2 are topological spaces). My original learning about quotient maps was fragmented so I never wrote that down before, but I think it's worth remembering in the future.

One last exercise I want to do before forcing myself to try to sleep is as follows:

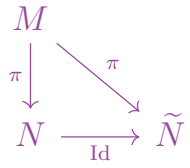
Exercise 4-7: Let M and N be C^r manifolds without boundaries, and let $\pi : M \to N$ be a surjective C^r submersion. Show there is no other manifold topology and C^r structure that we can equip N with such that for any C^r manifold P and map $F : N \to P$ we have that F is C^r if and only if $F \circ \pi$ is C^r.

Proof:
Let $\widetilde{N}$ be the same set as N except equipped with a potentially different manifold topology and C^r structure. Since the following diagram commutes and Id is a C^r map from $\widetilde{N}$ to $\widetilde{N}$,

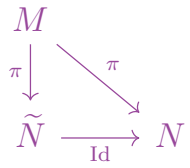
we can conclude that $\pi : M \to \widetilde{N}$ is a C^r map.



But then since $\pi : M \to \widetilde{N}$ is a C^r map we can also conclude from the following diagram that $\mathrm{Id} : N \to \widetilde{N}$ is a C^r map.



Meanwhile, as $\pi : M \to N$ is a C^r map we can conclude from the following diagram that $\mathrm{Id} : \widetilde{N} \to N$ is a C^r map.



So, we've proven that $\mathrm{Id} : N \to \widetilde{N}$ is a C^r-diffeomorphism. Since this also means Id is a homeomorphism, we can conclude that the topologies on N and $\widetilde{N}$ are identical. Also, $\mathrm{Id} : N \to \widetilde{N}$ being a C^r diffeomorphism easily lets us conclude that every smooth chart on N is C^r compatible with every smooth chart on $\widetilde{N}$ and vice versa.

Why?
Suppose (U, φ) is a smooth chart with respect to N and (V, ψ) is a smooth chart with respect to $\widetilde{N}$ such that $U \cap V \neq \emptyset$. Then as $\mathrm{Id} : N \to \widetilde{N}$ is a C^r-diffeomorphism, we know that $\psi \circ \varphi^{-1} = \psi \circ \mathrm{Id} \circ \varphi^{-1}$ and $\varphi \circ \psi^{-1} = \varphi \circ \mathrm{Id} \circ \psi^{-1}$ are C^r maps.

It follows that the C^r structures on N and $\widetilde{N}$ are also the same. ■

In particular, this shows that if M is a C^r manifold without a boundary and $\pi : M \to N$ is any surjective function then there is at most one manifold topology without boundary and C^r structure on N such that π is a C^r submersion.

Chapter 5 of John Lee's book will let us fully connect the abstract definition of a manifold with the more naive definition studied on *pages 77-99*.

Suppose M is a C^r manifold. A C^r embedded submanifold of M is a subset $S \subseteq M$ which is a topological manifold when equipped with the subspace topology and which has a C^r structure such that the inclusion map $\iota : S \to M$ is a C^r embedding.

Remarks:

- If $k \leq r$ then M has a unique C^k structure containing its C^r structure. Based on this structure on M, if S is a C^k manifold then we can ask if $\iota : S \to M$ is a C^k embedding. And if so we call S a C^k embedded submanifold of M.
- Some authors also refer to embedded submanifolds as regular submanifolds.

If M is a C^r manifold and $S \subseteq M$ is a C^k embedded submanifold of M then the codimension of S in M is defined to be the difference $\dim(M) - \dim(S)$. Here M is called the ambient manifold for S. If S has codimension 1 we say S is an embedded hypersurface.

Proposition 5.1: Suppose M is a C^r manifold without a boundary. Then the embedded C^k submanifolds of codimension 0 (without boundaries) in M are precisely the open submanifolds of M.

Proof:
If $U \subseteq M$ is an open submanfold then we know $\dim(U) = \dim(M)$. Furthermore, U is equipped with the subspace topology and we know from before that the inclusion map $\iota : U \hookrightarrow M$ is a C^k immersion and submersion. It follows that ι is a C^k embedding. And as M has no boundary, neither does U.

Conversely, suppose $U \subseteq M$ is an embedded C^k submanifold in M of codimension 0. By definition the inclusion map $\iota : U \hookrightarrow M$ is a C^k embedding. Yet as $\dim(M) = \dim(U)$ and ι is an immersion, we know from proposition 4.8(b) on *page 1061* that ι is a local C^k-diffeomorphism. In turn, by proposition 4.6(c) on *page 1060* we know ι is an open map. So, $\iota(U) = U$ is an open subset of M.

Finally, suppose (V, φ) is a smooth chart on M (equipped with its unique C^k structure). Then $U \cap V$ is an open subset of U and I claim that $(U \cap V, \varphi)$ is a smooth map on U. After all, suppose (W, ψ) is any other smooth chart on U as an embedded submanifold. Then as $\iota : U \hookrightarrow M$ is a C^k map we know that $\varphi \circ \iota \circ \psi^{-1} = \varphi \circ \psi^{-1}$ is C^k on $\psi((U \cap V) \cap W)$.

Meanwhile, as ι is a local C^k-diffeomorphism, we know for every $p \in (U \cap V) \cap W$ that there is some open set $\Omega \subseteq U$ containing p such that ι is a C^k-diffeomorphism from Ω (as an open submanifold of U) to Ω (as an open submanifold of M). In turn, $\psi \circ \varphi^{-1}$ is a C^k map on $\psi((U \cap V) \cap W \cap \Omega)$. So, $\psi \circ \varphi^{-1}$ is C^k at every $p \in (U \cap V) \cap W$. And with that, we've proven that the C^k structure on U as an embedded submanifold must be the same as the C^k structure on U as an open submanifold. ■

Proposition 5.2: Suppose M and N are C^r manifolds, and $F : N \to M$ is a C^r embedding. Let $S = F(N)$. With the subspace topology S is a topological manifold. Also S has a unique C^r structure making it into an embedded submanifold of M and satisfying that F is a C^r-diffeomorphism from N to S.

Proof:
S being a subspace of M and M being second countable and Hausdorff means that S is also second countable and Hausdorff. Also by assumption F is a homeomorphism from N to S. Therefore, N being locally Euclidean of dimension $\dim(N)$ means that S is as well.

Now we equip S with the C^r structure given by the charts $(F(U), \varphi \circ F^{-1})$ where (U, φ) is any smooth chart on N. Them being C^r compatible with each other is a result of the smooth charts on N being C^r compatible with each other. So, S is a C^r manifold. It's also easy to see now that F is a C^r-diffeomorphism. After all, for any $p \in S$ we can pick a smooth chart (U, φ) on N containing $F^{-1}(p)$. Then $(F(U), \varphi \circ F^{-1})$ is a smooth chart on S containing p, and:

$$\varphi \circ F \circ (\varphi \circ F^{-1})^{-1} = \mathrm{Id}_{\varphi(U)} \text{ is a } C^r \text{ map.}$$

So, $F^{-1} : S \to N$ is a C^r map.

Next, to show uniqueness just note that if F is to be a diffeomorphism from N to S then for any smooth chart (U, φ) on N we must have that $(F(U), \varphi \circ F^{-1})$ is a smooth chart on S. Similarly, for every smooth chart (V, ψ) on S we must have that $(F^{-1}(V), \psi \circ F)$ is a smooth chart on N. So the atlas described in the prior paragraph is maximal and F being a C^r-diffeomorphism from N to S automatically implies that S is equipped with that maximal structure.

Finally, I need to show that the map $\iota : S \to M$ is a C^r-embedding. Fortunately, consider that ι is given by the composition:

$$S \xrightarrow{F^{-1}} N \xrightarrow{F} M \qquad (\iota : S \to M)$$

Since F^{-1} is a C^r-diffeomorphism and F is a C^r-embedding, we can conclude ι is a C^r embedding. So, S is an embedded submanifold of M. ■

> As a side note, if $F : N \to M$ is a C^k embedding (where $k \leq r$) we can also now say that $S = F(N)$ has a unique C^k structure making it into an embedded submanifold of M such that F is a C^k-diffeomorphism from N to S. After all, we can just re-equip M and N with the unique C^k structures containing their original C^r structures and then everything works the same.

Proposition 5.3: Suppose M and N are C^r manifolds. For each $p \in N$ the subset $M \times \{p\} \subseteq M \times N$ (called a slice of the product manifold) is an embedded submanifold of $M \times N$ diffeomorphic to M.

Proof:
The map $F : M \to M \times N$ given by $F(m) := (m, p)$ is a C^r embedding. After all, it's a C^r map that is a topological embedding. Furthermore, recall from *page 1059* how we standardly identify $T_{(m,p)}(M \times N)$ with $T_m M \oplus T_p N$.

Lemma: Suppose $N, M_1, \ldots, M_k$ are C^r manifolds. Then let $M := \prod_{i=1}^k M_i$ and for each i let $\pi_i : M \to M_i$ be the projection map. Given any $(p_1, \ldots, p_k) \in M$ we identify $T_{(p_1,\ldots,p_k)} M$ with $\bigoplus_{i=1}^k T_{p_i} M_i$ via the standard identification. Now suppose for each $1 \leq i \leq k$ that $F_i : N \to M_i$ is a C^1 map. Then let $F : N \to M$ be defined by $F(q) = (F_1(q), \ldots, F_k(q))$. Now for any $q \in N$ we have that:

$$\mathrm{d}F_q(v) = (\mathrm{d}(F_1)_q(v), \ \ldots, \ \mathrm{d}(F_k)_q(v)).$$

Why?
Suppose $v \in T_qN$ and $f : M \to \mathbb{R}$ is a C^1 map. Then $\mathrm{d}F_q(v)f$ is identified with:

$$(\mathrm{d}(\pi_1)_{F(q)} \circ \mathrm{d}F_q(v), \ldots \mathrm{d}(\pi_k)_{F(q)} \circ \mathrm{d}F_q(v)) = (\mathrm{d}(\pi_1 \circ F)_q(v), \ldots, \mathrm{d}(\pi_k \circ F)_q(v))$$
$$= (\mathrm{d}(F_1)_q(v), \ \ldots, \ \mathrm{d}(F_k)_q(v))$$

Based on the prior lemma, we can see for any $v \in T_mM$ that:

$$\mathrm{d}F_m(v) = (v, 0) \in T_mM \oplus T_pN.$$

It follows that $\mathrm{d}F_m$ is injective for all $m \in M$. So, F is an immersion. ■

Proposition 5.4: Suppose M is a C^r m-dimensional manifold, N is a C^r n-dimensional manifold, $U \subseteq M$ is open, and $f : U \to N$ is a C^k map. Let $\Gamma(f) \subseteq M \times N$ denote the graph of f:

$$\Gamma(f) \coloneqq \{(m, f(M)) \in M \times N : m \in U\}$$

Then $\Gamma(f)$ is a C^k m-dimensional embedded submanifold of $M \times N$.

Proof:
Define a map $\gamma_f : U \to M \times N$ by $\gamma_f(m) = (m, f(m))$. This will be a C^k map because the inclusion map $\iota : U \to M$ is C^r and similarly the map $f : U \to N$ is C^k. Also let $\pi_M : M \times N \to M$ be the projection map. Then $\mathrm{Id}_U = \pi_M \circ \gamma_f$ implies that γ_f injective. Also for any $m \in U$ we have $\mathrm{d}(\pi_M)_{(m,f(m))} \circ \mathrm{d}(\gamma_f)_m = \mathrm{d}(\mathrm{Id}_U)_m = \mathrm{Id}_{T_mU}$. So, for every $m \in U$ we have that $\mathrm{d}(\gamma_f)_m$ is injective. In other words, γ_f is an immersion. Finally, γ_f is a topological embedding since its inverse function is given by $\pi_M|_{\Gamma(f)}$.

And now we've shown that $\Gamma(f)$ is the image of a C^k embedding γ_f from U. ■

An embedded submanifold $S \subseteq M$ is said to be properly embedded if the inclusion map $S \hookrightarrow M$ is a proper map.

Proposition 5.5: Suppose M is a C^r manifold and $S \subseteq M$ is an embedded submanifold. S is properly embedded if and only if S is a closed subset of M.

Proof:
Suppose the inclusion map $\iota : S \to M$ is a proper map. Then we know that it is a closed map (see *page 1104*). In turn, $S = \iota(S)$ must be a closed subset of M.

Conversely, note that ι being a topological embedding with a closed image implies that ι is proper.

This is because if $K \subseteq M$ is a compact set, then $S \cap K$ will also be compact since S is closed. And now as ι^{-1} is a continuous map, we know that $\iota^{-1}(K) = \iota^{-1}(S \cap K)$ is compact. So, ι is proper. ■

Corollary 5.6: Every compact embedded submanifold is properly embedded.

Proof:
Compact subsets of Haudorff spaces are closed. ■

Proposition 5.7: Suppose M and N are C^r manifolds, and $f : M \to N$ is a C^k map. Then after equipping $\Gamma(f) \coloneqq \{(m, f(M)) \in M \times N : m \in M\}$ with the smooth structure defined in proposition 5.4, we have that $\Gamma(f)$ is properly embedded.

Proof:
Note that $\gamma_f : M \to M \times N$ given by $\gamma_f(m) = (m, f(m))$ has the map $\pi_M : M \times N \to M$ given by $\pi_M(m, n) = m$ as a continuous left inverse. In other words, $\pi_M \circ \gamma_f = \mathrm{Id}_M$. As a result we can conclude that $\gamma_f : M \to M \times N$ is a proper map. To see why, I claim the following:

Lemma: If X, Y are topological spaces and $g : X \to Y$ and $h : Y \to X$ are continuous functions such that $h \circ g = \mathrm{Id}_X$, then g must be proper.

Why?
Suppose $K \subseteq Y$ is a compact set. Since h is continuous, we know that $h(K) \subseteq X$ is a compact set. But also let $F := g^{-1}(K)$. Then:

$$\begin{aligned} F = \mathrm{Id}_X(F) = (h \circ g)(F) &= h(K \cap g(X)) \\ &= h(K) \cap (h \circ g)(X) \\ &= h(K) \cap \mathrm{Id}_X(X) = h(K) \cap X = h(K). \end{aligned}$$

So, we've shown that $g^{-1}(K)$ is compact.

Since γ_f is a continuous proper map though and $M \times N$ is an LCH space, we can conclude that γ_f is a closed map. So $\gamma_f(M) = \Gamma(f)$ is a closed subset of $M \times N$. And by proposition 5.5, we are done. ■

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